Fine-Tuning & Distillation

LoRA + knowledge distillation, from scratch

October 9, 2025

Prepared for: Sai Veda
Publishing account: Nikeshk834

Headline Result
LoRA 91% params; KD recovers 100% of teacher

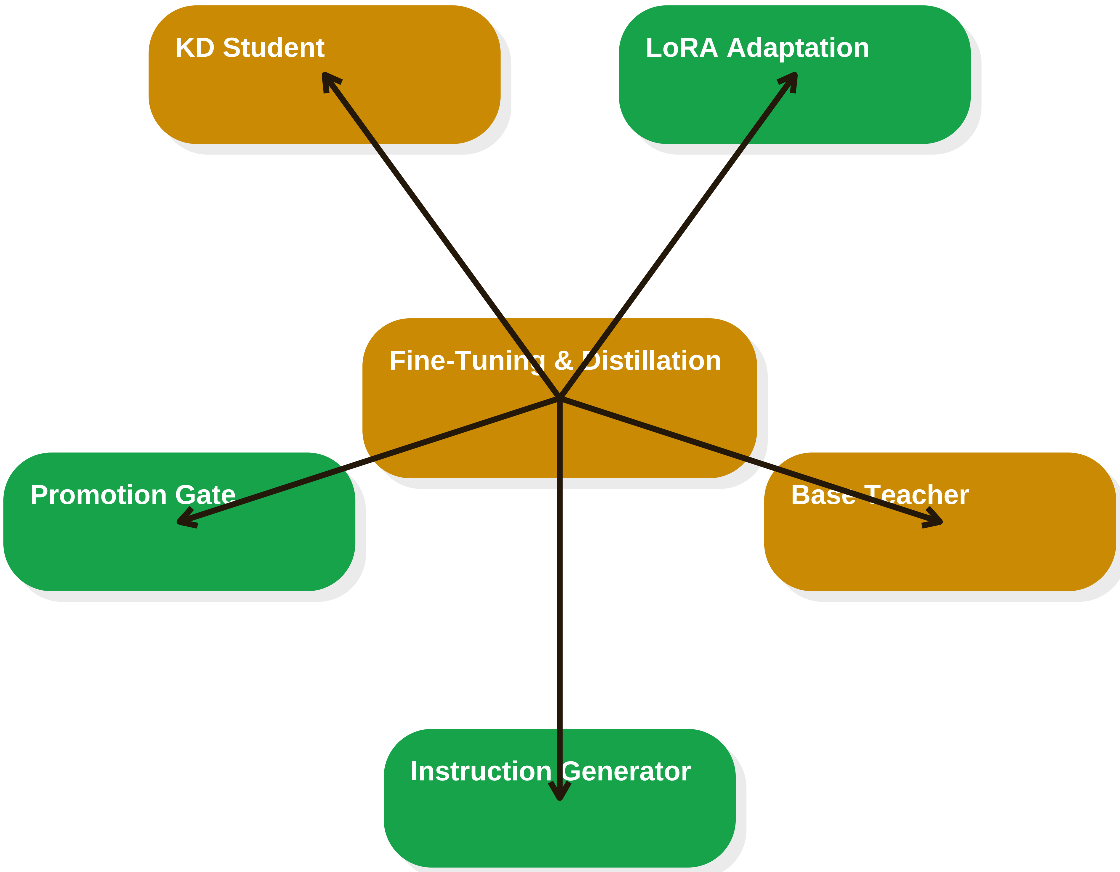
Project Context
A fully-runnable, GPU-free pipeline that implements the core techniques used to adapt and compress large language models - full fine-tuning, LoRA low-rank adaptation, knowledge distillation, an automated eval/promotion gate, and a synthetic-data scaling study - end to end, deterministically, on a CPU in a few minutes.

This PDF dossier is intentionally visual and interview-ready: it captures the business problem, the technical architecture, the validation evidence, and the committed repository artifacts in a format that reads like a delivered project brief.

Repository slug	08-llm-finetuning-distillation	**Validation index**	29 tests

Architecture Storyboard

This flow mirrors a real adaptation stack: a stronger teacher, a parameter-efficient adapter, and a hard promotion gate.



High-level focus area: This flow mirrors a real adaptation stack: a stronger teacher, a parameter-efficient adapter, and a hard promotion gate.

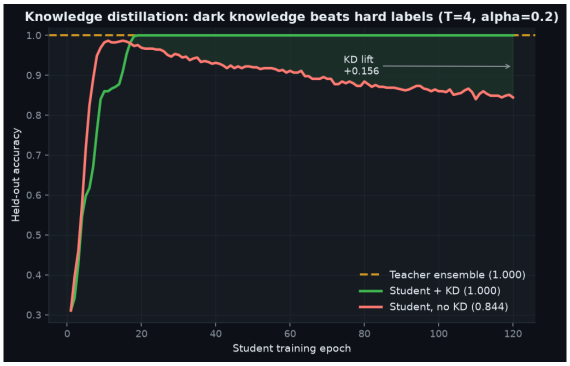
Validation And Visual Evidence

LoRA 91% params; KD recovers 100% of teacher

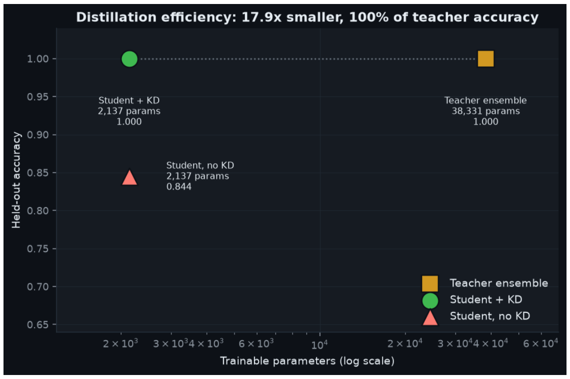
29 passing tests **October 9, 2025**

Selected benchmark table

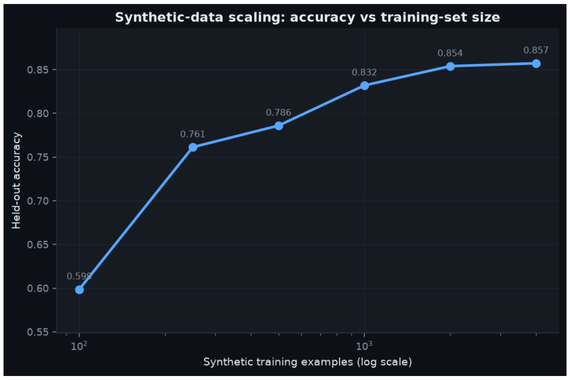
Method	Target-domain accuracy	Trainable params
Full fine-tune	0.934	33,417
LoRA (r=4)	0.934	2,889



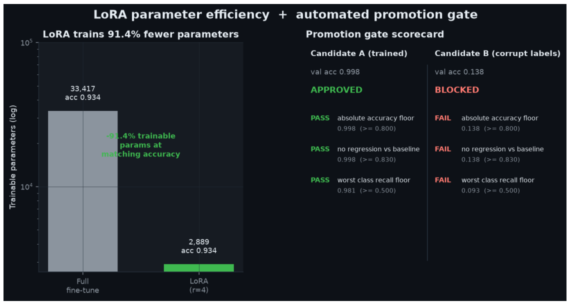
Knowledge distillation: dark knowledge beats hard labels



Distillation efficiency (parameters vs accuracy)



Synthetic-data scaling



LoRA parameter efficiency + promotion-gate scorecard

Repository Handoff

What is committed, where the proof lives, and how this project should be reviewed.

Prepared for: Sai Veda
Publishing account: Nikeshk834
Project date: October 9, 2025
Repository: 08-llm-finetuning-distillation

A fully-runnable, GPU-free pipeline that implements the core techniques used to adapt and compress large language models - full fine-tuning, LoRA low-rank adaptation, knowledge distillation, an automated eval/promotion gate, and a synthetic-data scaling study - end to end, deterministically, on a CPU in a few minutes. All numbers below are produced by python scripts/run_pipeline.py and stored in benchmarks/results.json / benchmarks/results.md.

Committed deliverables

- README.md - product overview, quickstart, and measured headline numbers
- ARCHITECTURE.md - detailed design and scale trade-offs
- tests/ - automated validation suite
- benchmarks/ - measured run artifacts and result tables
- assets/ - generated dashboards, charts, and screenshots
- PROJECT_DOCUMENT.pdf - styled project brief for sharing and review

This document is intentionally more visual than the repository Markdown. It is meant to read like a polished project brief: business framing first, architecture second, proof third, and delivery scope last.